Multiscale and Dense Ship Detection in SAR Images Based on Key-Point Estimation and Attention Mechanism

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***Abstract*— Ship target detection in synthetic aperture radar (SAR) images is essential for many applications in marine monitoring and port security. Though considerable developments have been achieved, there still exist some issues toward multiscale and dense ship targets in complex inshore scenes. Under such common but challenging situations, it is difficult to extract effective target information, which drives the missing alarm rate rising dramatically. In complex scenes, it is hard to disentangle background noise from ship target information, which causes false alarm frequently. In this article, an anchor-free SAR ship detection method based on key-point estimation and attention mechanism is proposed to address the aforementioned issues. Specially, an anchor-free framework with skip connections and aggregation nodes is designed to fuse multiresolution features and detect multiscale ship targets. Moreover, a key-point estimation module is proposed to eliminate the undetected ship targets caused by dense target distribution. Furthermore, a channel attention module is explored to enhance network attention on ship targets and suppress background noise. Sufficient experimental results on the open SAR ship detection dataset demonstrate that compared with some state-of-the-art methods, the proposed method is able to achieve higher detection accuracy with a lower false alarm rate.**

***Index Terms*— Multiscale target, ship detection, synthetic aperture radar (SAR) image.**

I. Introduction

SINCE synthetic aperture radar (SAR) images, which are not limited by light intensity or weather conditions, exhibit better robustness than natural optical images in terms of visual and statistical features. Therefore, SAR images are extensively applied in earth observation tasks [1], particularly in marine target monitoring [2], maritime rescue [3], and port security [4]. Because of the specialty of SAR images, ship target detection using SAR images has clear advantages, which make large-coverage all-weather ship surveillance task possible.

In the past decades, a lot of methods have been proposed in SAR images ship target detection [5]–[10]. Among all traditional detection methods, constant false-alarm rate (CFAR)-based method is the most widespread one. Wang *et al.* [11] proposed an intensity-space domain CFAR detector, which exploits the data correlation to detect target pixels and wake pixels. Leng *et al.* [12] proposed a CFAR detector with bilateral structure, which reduces the disturbance of speckle noise and sea clutter by considering both the intensity information and locality information. Pappas *et al.* [13] proposed a superpixel-level CFAR detector, which provides superior performance in reducing false alarm by utilizing the superpixel technique to obtain regions of interest. These methods have achieved competitive performance by analyzing the characteristics of SAR images when the scenes are relatively simple. However, when target regions occupied by few pixels, which is a common situation in SAR images, CFAR-based methods are hard to extract effective target information, which leads to a limited performance with high missing alarm rate.

Recently, as the booming of deep learning theory, SAR images ship target detection methods based on deep network have made great progress, especially in effective feature extraction. To improve detection accuracy and efficiency, many studies have been put forward and achieved good performance in various aspects. To solve the issue of multiscale targets, some researches extract and fuse multilevel features. Jiao *et al.* [14] designed a network with densely connected structure, which aims at address multiscale problem by merging high resolution feature maps in upper level layer with feature maps extracted from lower level layer. Fu *et al.* [15] proposed a feature balancing and refinement network, which achieves a great improvement in detection performance by refining the target features and strengthening semantic information. Cui *et al.* [16] proposed a ship detection method in large-scale SAR images via CenterNet, which applied an attention module with group convolution to enhance learned the features and exclude the interference of the coast on ship target detection. Du *et al.* [17] proposed a novel single shot multibox detector with salient guidance, which could focus more on ship targets by introducing the saliency information into the

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network. Some other works try to improve the detection speed, and propose lightweight models for feature extraction [18], [19]. In the normal case, the aforementioned methods have improved the performance of ship detection with more intelligent learning methods. However, for real applications, there are some issues that still need to be addressed. For inshore area, ship targets with multiscale sizes anchor densely. It is difficult to extract effective target information in practical applications, which leads to serious false alarms and missed detection. Therefore, for the purpose of advancing practical SAR images ship target detection, how to detect multiscale and dense ship targets have to be settled urgently.

To improve detection performance of multiscale and dense ship targets in complex scenes, this article considers from the following aspects, and proposes an anchor-free method based on key-point estimation and attention mechanism. Firstly, the proposed method works under an anchor-free framework, which is more adaptable for multiscale targets. Specially, unlike other anchor-based target detection methods, which enumerate a nearly exhaustive list of potential target locations and classify each, the proposed method finds the center points instead of setting the anchors artificially. Secondly, the proposed method fuses multiresolution feature maps through skip connections and aggregation nodes, which can combine and compress the feature maps by learning to select and represent important information, and the residual connections are essential for utilizing spatial information. Thirdly, compared with the output resolution of other target detection methods, the proposed method utilizes a larger output resolution to extract features and locate the targets with small size effectively. Fourthly, when the ship targets lie alongside and connect densely in the inshore area, the center points of different targets stay closely even overlap others. Detectors are hard to match each point to its corresponding target. Therefore, the proposed method designs a key-point estimation module to address this issue. Lastly, the proposed method explores a channel attention module to make the network focus on ship targets and eliminate the interference of background noise. The main contributions of our work are summarized as follows.

1) We propose an anchor-free ship detection method based on a learning network, it models each target by a single point and regresses to the size of ship targets with skip connections and aggregation nodes, which can effectively detect multiscale ship targets.
2) We design a key-point estimation module, it extracts the peak key-points from the heat-map using three connected neighbors, which can locate the center point of target precisely and reduce the missing alarm in dense targets area.
3) We design a channel attention module, it utilizes the gating mechanism to make full use of channel-wise statistics, which enhances the power of extracting effective target information and eliminates the false alarms caused by background noise.

The organization of the following sections is introduced here. Section II briefly introduces the related works on SAR images ship target detection. Section III gives a detailed description on the implementation of our proposed anchor-free SAR ship detection method. Section IV specifically illustrates the experiments, experimental settings, experimental results, and corresponding experimental analysis. Section V concludes the performance of our proposed method and puts forward the prospect of the future works.

II. Related Works

There are two main types of SAR images ship target detection methods, i.e., traditional methods, and deep learning methods. The traditional methods are mainly divided into three categories, i.e., statistical characteristics algorithm [20], threshold algorithm, and transformation algorithm [21]. Artificial feature extraction is a remarkable characteristic of traditional methods, e.g., gray-level, shape, contrast ratio, histogram of oriented gradient [22], scale-invariant feature transform [23], Haar-like [24], and other features. CFAR-based methods are the most representative traditional methods, which utilize the gray-scale feature and achieve good performance in simple scenes. However, when the scene is complex in practice, such as the shore, harbor, and port, an applicative clutter statistical distribution is difficult to model, which make the detection performance drop drastically. What's worse, the severe coherent speckle noise also challenges SAR images ship target detection, and makes it harder for traditional SAR ship methods to detect ship targets.

With the rapid development of deep learning in target detection, the detection performance has been improved to an unprecedented high level. Different from traditional methods, the deep learning target detection methods can extract features automatically, which provides effective representation ability. There are two main target detection methods based on deep learning. One is single-stage target detection method, such as you only look once (YOLO) [25], and single-shot multibox detector (SSD) [26], which spends less time to detect. But its detection performance in multiscale targets drops dramatically for the large receptive field of the last convolutional layer. The other one is two-stage target detection method, which proposes candidate target bounding boxes independent of the class at first, and then classifies them, such as the region-convolutional neural network (R-CNN) [27]. However, R-CNN is accompanied by large quantities of calculations, which causes a very limited detection speed. After that, Fast R-CNN is proposed to solve this problem by obtaining a region of interest (ROI) in feature maps. Faster R-CNN [28] is proposed to improve the detection speed by designing a region proposal network (RPN). For the aforementioned methods, spatial information of the target has lost severely in the process of convolution. To combine multilevel features, the feature pyramid network (FPN) [29] is proposed to extract features from different layers to strengthen the ability of location and classification. Non-maxima suppression (NMS) [30] is proposed to remove repetitive bounding boxes based on the intersection over union (IoU) threshold. However, for SAR images, when ship targets lie alongside and connect densely in the shore area, the aforementioned detection methods are difficult to detect all ship targets because of the artificial anchor boxes size and NMS threshold constraint.

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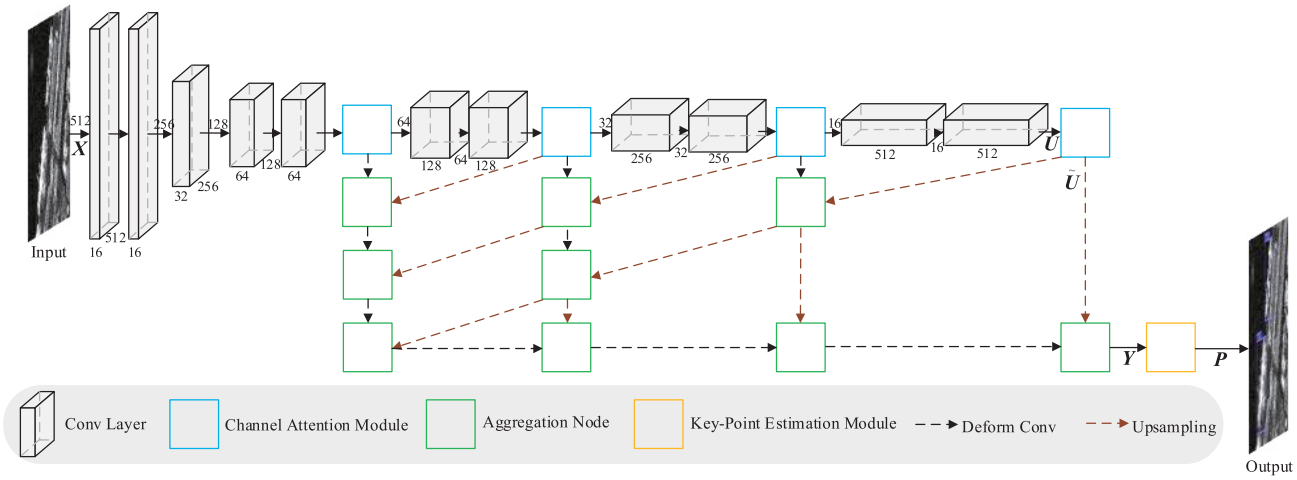


Fig. 1. Overview of the proposed anchor-free target detection method. The key-point estimation module and channel attention module, which will be introduced detail in the following sections, are shown in the framework. Skip connections from the bottom layers with deformable convolution and upsampling are adopted to combine multiresolution in the proposed method. $\boldsymbol{X}$ is the input SAR image, $\boldsymbol{U}$ is the extracted feature map, $\tilde{\boldsymbol{U}}$ is the enhanced feature map, $\boldsymbol{Y}$ is the generated key-point heat-map, and $\boldsymbol{P}$ is the peak key-points.

Recently, the anchor-free target detection methods have attracted lots of attention. They have achieved excellent performance, especially in the targets with multiscale sizes. There are mainly two categories of anchor-free detection method, i.e., anchor-point estimation [31], and key-point estimation [32]. CornerNet [33] models two bounding box corners as key-points, which first utilizes the key-point estimation to detect the target and completely abandons the concept of anchor. ExtremeNet [34] detects the top, left, bottom, right, and center points of the target, which utilizes the state-of-the-art key-point estimation to obtain representative points of each target. The aforementioned methods all contain combination and grouping process after key-point estimation, which brings uncertain grouping condition and large time cost. Unlike the aforementioned methods, CenterNet [35] simply detects the center point of each target bounding box, and has remarkable adaptability of target size because of its small down-sampling factor. Thus, CenterNet has promotions in the aforementioned issues, and the performance is improved both on accuracy and speed. Because CenterNet cannot match each center point to its corresponding ship target precisely and disentangle the background information from ship targets efficiently, when it performs in SAR images, there are many false alarms and undetected targets. Therefore, to address the aforementioned issues, we propose an anchor-free detection method and design a key-point estimation module to locate the center point of target precisely and reduce the false alarm and missing alarm.

For feature extraction, deep convolutional neural networks, e.g., VGGNets [36], and Inception models [37], have revealed that the depth of a network can effectively improve the ability to learn the representative features of targets. But with the depth of networks increasing, the complexity and design difficulty of the network are harder. To obtain more representational and robust networks, up-convolutional residual networks (ResNet) [38] utilizes the identity-based skip connections, which extracts more representative features, but introduces background information and noise at the same time. In this case, lots of works abandon manual networks design and learn the structure of networks automatically. Among them, squeeze-and-excitation (SE) block [39] is of high efficiency and the most widely used. SE block consists of a gating mechanism which improves the learnable and representative ability by adding channel-wise relationships. Therefore, to disentangle background information from ship targets and highlight representative target features, we propose an anchor-free detection method with SE block and design a channel attention module to enhancing the attention of network toward ship targets.

III. Methodology

In this section, we give a detailed introduction of the proposed method. The overall architecture of the proposed method is shown in Fig. 1. There are four key structures in the proposed deep architecture, fully convolutional layers are utilized to extracted the features of the input SAR image, new designed channel attention modules are combined with the network to suppress the background noise and highlight the target features, aggregation nodes are used to fuse multiresolution feature maps and represent important information, a key-point estimation module is proposed to generate the key-point heat-map and enhance the detection rate of densely connected targets. The detailed information about the anchor-free framework and the proposed modules are presented in Sections III-A–C. The descriptions of main notations we used are shown in Table I.

A. Anchor-Free Framework

We develop an anchor-free framework based on CenterNet, which detects the center point of the bounding box for each target by key-point estimation and regresses to get all other target properties, e.g., target size.

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TABLE I
NOTATION DEFINITIONS

Notation	Description
$\boldsymbol{X}$	The input SAR image
$\boldsymbol{Y}$	The generated key-point heat-map
$\boldsymbol{Z}$	The ground truth key-point heat-map
$\boldsymbol{O}$	The local offset
$\boldsymbol{S}$	The size of target
$\boldsymbol{P}$	The peak key-points
$\boldsymbol{U}$	The extracted feature map
t	The channel-wise statistics
w	The weights of channels
$\tilde{\boldsymbol{U}}$	The enhanced feature map

Supposed $\boldsymbol{X} \in \mathbb{R}^{H \times W}$ is the input SAR image of height H and width W, $\boldsymbol{Y} \in [0,1]^{(H/R)\times(W/R)}$ is the generated key-point heat-map of downsampling factor R, $\boldsymbol{Z} \in [0,1]^{(H/R)\times(W/R)}$ is the ground truth key-point heat-map established by a Gaussian kernel

$$\boldsymbol{Z}(x,y) = \exp\left(-\frac{\left(x-d'_x\right)^2 + \left(y-d'_y\right)^2}{2\sigma_d^2}\right) \tag{1}$$

where $d' = \lfloor (d/R) \rfloor$ corresponds to the key-points locations d after downsampled by the factor R, σ_d is a standard deviation that used to control its distribution and (x, y) is the coordinate position in the range of height (H/R) and width (W/R).

An example of the ground truth key-point heat-map $\boldsymbol{Z}$ is shown in Fig. 2. The center point value of the ship target bounding box is set as 1. Value of other points in the bounding box are all less than 1 and the farther away from the center point, the lower they are.

At the beginning, the proposed method feeds $\boldsymbol{X}$ to multiple fully convolutional layers. The ideal output result of the network is a downsampled key-point heat-map $\boldsymbol{Y}$ with factor R, where the center of target is predicted to be 1 and the background is predicted to be 0. Since focal loss can solve the sample imbalance problem by sample mining [40], it is utilized as loss function to train the proposed anchor-free network through (2), as shown at the bottom of the page, α and β are hyper-parameters of the focal loss, N is the number of key-points in input image $\boldsymbol{X}$, $\boldsymbol{Y}$ is the key-point estimation heat-map result, $\boldsymbol{Z}$ is the ground truth key-point heat-map, and (x, y) is the coordinate position in the range of height (H/R) and width (W/R).

Then, the local offset $\boldsymbol{O} \in \mathbb{R}^{(H/R)\times(W/R)\times 2}$ is proposed to eliminate the discretization error caused by downsampling [35]. $\boldsymbol{O}$ represents the direction information between the key-point and the corresponding predicted positions within a certain range from the key-point, i.e., the distance between the key-point and all positions with a prediction of 1. Particularly, $\boldsymbol{O}$ only acts at the key-points locations d', and defined by an L1 loss function as follows:

$$L_{\text{off}} = \frac{1}{N}\sum_{d}\left|\boldsymbol{O}_{d'} - \left(\frac{d}{R} - d'\right)\right|. \tag{3}$$

In addition, the size $\boldsymbol{S} \in \mathbb{R}^{(H/R)\times(W/R)\times 2}$ of predicted target are regressed at the center point, which is also defined by an L1 loss function as follows:

$$L_{\text{size}} = \frac{1}{N}\sum_{i=1}^{N}\left|\boldsymbol{S}_{d_i} - \boldsymbol{s}_i\right| \tag{4}$$

where d_i denotes the ith key-point location of d and $\boldsymbol{s}_i$ denotes the size of each target i.

At last, we combined the aforementioned three loss functions to get an optimal effect. The overall objective function is defined as

$$L_{\text{det}} = L_k + \lambda_{\text{off}} L_{\text{off}} + \lambda_{\text{size}} L_{\text{size}} \tag{5}$$

where λ_{off} and λ_{size} are hyper-parameters of the overall loss. They are used to balance the proportion of key-point estimation loss, offset regression loss, and size regression loss.

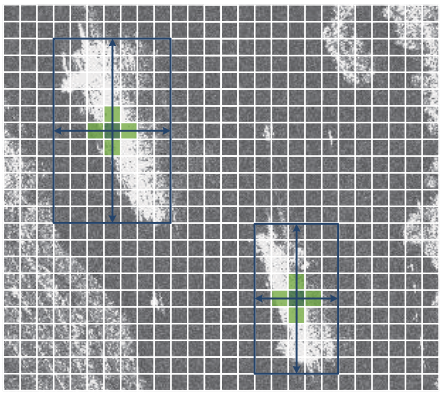


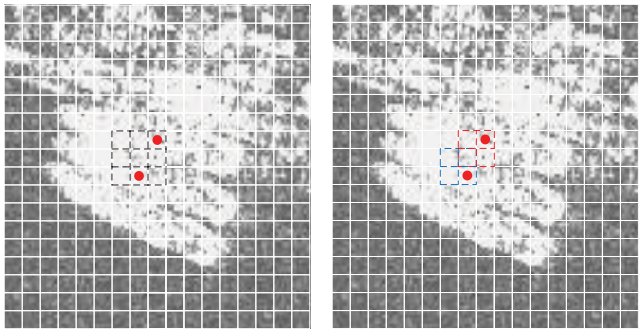
Fig. 2. Ground truth key-point heat-map established by the Gaussian kernel function.

B. Key-Point Estimation Module

We propose a novel key-point estimation module. Firstly, the key-point heat-map $\boldsymbol{Y}$ is generated by the fully convolutional network. Then, our purpose is to extract the peak key-points $\boldsymbol{P} \in [0,1]^{(H/R)\times(W/R)}$ of $\boldsymbol{Y}$. By comparing the pixel values with other predicted results that are their connected neighbors, we only preserve the maximum pixel values while deleting other pixel values. In this process, we give up using its 8−connected neighbors (top, bottom, left, right, top-left, top-right, bottom-left, and bottom-right), as shown in Fig. 3(a), but

$$L_k = -\frac{1}{N}\sum_{x=1}^{\frac{H}{R}}\sum_{y=1}^{\frac{W}{R}}\begin{cases}[1-\boldsymbol{Y}(x,y)]^{\alpha}\log \boldsymbol{Y}(x,y), & \text{if } \boldsymbol{Z}(x,y)=1\\ [1-\boldsymbol{Z}(x,y)]^{\beta}[\boldsymbol{Y}(x,y)]^{\alpha}\log[1-\boldsymbol{Y}(x,y)], & \text{otherwise}\end{cases} \tag{2}$$

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(a) (b)

Fig. 3. Mode of extracting the peak key-points from the heat-map. (a) Original eight-connected neighbors' mode. (b) Three-connected neighbors mode of the key-point estimation module. The red points denote that the pixels exist center points of targets.

its 3−connected neighbors (bottom, right, and bottom-right), as shown in Fig. 3(b).

In detail, assume that the key-point heat-map $\boldsymbol{Y}$ of the input image $\boldsymbol{X}$ has been obtained. We extract the peak key-points $\boldsymbol{P}$ through a 2×2 max pooling operation. Then we set the pixel values of $\boldsymbol{P}$ that are not peak key-point to 0. Combining $\boldsymbol{P}$ with the local offset $\boldsymbol{O}$ and size $\boldsymbol{S}$, we produce bounding boxes at each location (x, y) of the peak key-points remained in $\boldsymbol{P}$ through the following:

$$(x + \delta x - w/2, y + \delta y - h/2x + \delta x + w/2, y + \delta y + h/2) \tag{6}$$

where $(\delta x, \delta y) \in \boldsymbol{O}$ is the prediction of local offset, and $(w, h) \in \boldsymbol{S}$ is the prediction of target size.

C. Channel Attention Module

We explore a channel attention module to highlight the features of ship targets and enhance the learning and representative ability of the network. The structure of the channel attention module is shown in Fig. 4.

Supposed $\boldsymbol{U} \in \mathbb{R}^{H' \times W' \times C'}$ with height H', width W', and channel C' is the extracted feature map of the input image $\boldsymbol{X}$. The channel relationships are modeled by convolution. But they are specifically implicit and regional. We aim to enhance the power of ship targets features learning in the convolution operator. So it is advisable to increase the channel-wise sensitivity to the information of ship targets features.

Firstly, the channel-wise statistics $t \in \mathbb{R}^{C'}$ is acquired by a global average pooling through the following:

$$t_c = \frac{1}{H' \times W'} \sum_{x=1}^{H'} \sum_{y=1}^{W'} \boldsymbol{u}_c(x, y) \tag{7}$$

where t_c is the cth element of t and $\boldsymbol{u}_c \in \mathbb{R}^{H' \times W'}$ is the feature map of channel c in $\boldsymbol{U}$.

To obtain more representative channel-wise statistics t of the operation in (7), the gating mechanism is explored to capture the dependencies of channel-wise. We acquire the weights of channels $w \in \mathbb{R}^{C'}$ through (8). What's more, the fully

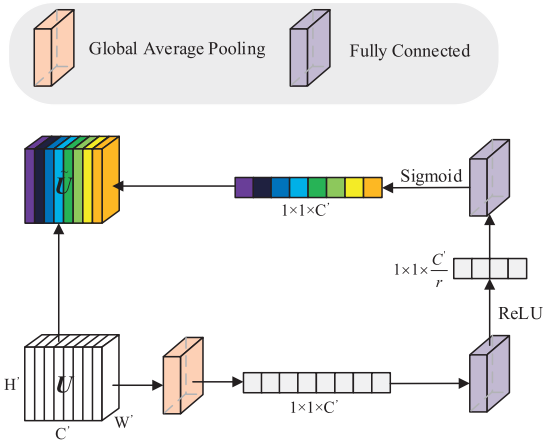


Fig. 4. Structure of the channel attention module.

connected operation is utilized to restrain the complexity and generalization of the channel attention module

$$w = \sigma(F_2 \delta(F_1 t)) \tag{8}$$

where σ denotes the sigmoid activation function, δ denotes the ReLU function, $F_1 \in \mathbb{R}^{(C/r) \times C}$, $F_2 \in \mathbb{R}^{C \times (C/r)}$ represent the fully connected operation, and r denotes the reduction ratio of the fully connected operation.

Finally, the enhanced feature map $\tilde{\boldsymbol{U}}$ after adding attention mechanism is gained through the following:

$$\tilde{\boldsymbol{u}}_c = w_c \boldsymbol{u}_c \tag{9}$$

where the scalar w_c is the c-th element of w, $\tilde{\boldsymbol{u}}_c \in \mathbb{R}^{H' \times W'}$ is the feature map of channel c in $\tilde{\boldsymbol{U}}$.

IV. Experiments

To evaluate the performance of the proposed anchor-free ship detection method, some experiments are implemented on the public SAR ship detection dataset. Firstly, we give detailed experimental settings of the proposed anchor-free ship detection method. Then, we verify the effectiveness of the key-point estimation module and the channel attention module. Finally, we compare the detection performance with other state-of-the-art detectors. Specially, our hardware platform is a desk computer with Intel Core i7 4.0-GHz CPU, NVIDIA GeForce RTX 2080Ti GPU, and 32-GB memory. The deep learning framework is based on PyTorch 1.4.0 with Python 3.6 and the operating system is Ubuntu 18.04.

A. Experimental Settings

1) Dataset: We implement the proposed anchor-free ship detection method on SAR Ship Detection Dataset (SSDD) [41] and SAR-ship-dataset [42]. All input of the proposed anchor-free ship detection method in our experiments are resized into 512×512 pixels.

There are 1160 SAR images with 2456 ships in SSDD captured by RadarSat-2, TerraSAR-X, and Sentinel-1 in Yantai, China, and Visakhapatnam, India. The average number of ships per image is 2.12. SAR images in SSDD include HH, HV, VV, and VH polarization modes. Its resolutions

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TABLE II
HYPERPARAMETERS SETTINGS

Hyper-parameters	Description	Setting
R	The upsampling factor	4
α	The penalty coefficient of prediction	2
β	The penalty coefficient of sample	4
λ_{size}	The coefficient of L_{size}	0.1
λ_{off}	The coefficient of L_{off}	1

range from 1 to 15 m. The size of each image sample is about 500 × 500. We select the images with the suffix of indexes 1 and 9 as the test set (232 images). The images with the suffix of indexes 7 are set as the validation set (116 images). The remaining images in SSDD are set as the training set (812 images).

SAR-ship-dataset is an open SAR ship dataset that published by the Chinese Academy of Sciences. There are 43819 SAR images with 59535 ships in SAR-ship-dataset captured by Gaofen-3 and Sentinel-1. The average number of ships per image is 1.36. The size of each image sample is 256 × 256. There are multiscales for the ships in the same image and the same ship in different resolution images. It is randomly divided into training set, verification set, and test set in an 8:1:1 ratio. In detail, there are 35055 images for training, 4382 images for verification, and 4382 images for test, respectively.

2) Experimental Settings: The settings of hyper-parameters are shown in Table II. In detail, a series of experiments with different factor R are conducted and finally the default output stride of R is set to 4 as in [43]–[45]. For parameters of the loss function, the hyper-parameters α and β of the focal loss in (2) are set to 2 and 4, respectively. To prevent the prediction $\boldsymbol{Y}(x, y)$ from being too high, i.e., closing to 1, $(\boldsymbol{Y}(x, y))^{\alpha}$ is used to punish the loss. The larger α, the greater degree of punishment for the prediction $\boldsymbol{Y}(x, y)$. $(1 - \boldsymbol{Z}(x, y))^{\beta}$ is used to reduce the punishment. The larger β, the smaller degree of punishment for the sample $\boldsymbol{Z}(x, y)$. We test with different values to determine α and β. The hyper-parameters λ_{size} and λ_{off} of the overall loss in (5) are set to 0.1 and 1. Since the calculation value of L_{size} is much larger than that of L_k and L_{off}, a smaller λ_{size} is designed to balance the overall loss. After several groups of experiments, we set λ_{size} and λ_{off} to converge the overall loss function to the global optimal solution.

For the backbone selection, several different fully convolutional encoder–decoder networks are studied as the backbone of our proposed method: A stacked hourglass network (Hourglass-104) [44], ResNet-50, and deep layer aggregation (DLA-34) [46]. DLA-34 combine and compress the feature maps through skip connections and aggregation nodes, which is able to detect multiscale ship targets effectively, especially those of small scale. Moreover, the network structure of DLA-34 is relatively uncomplicated, which has advantages in training time and detection speed. Therefore, DLA-34 is used as the backbone of our proposed anchor-free ship detection method.

For the training parameters, the batch size is set to 8, and the initial learning rate is set as 1.25e-4, which is then divided by a factor of 10 at the 100th and 150th epoch, respectively, the weight decay of 0.0001 and the momentum of 0.9. The training is ended with 200 epochs and attain the best model in the 170th epoch. SAR image differs from the traditional optical image in many aspects, e.g., resolution, speckle noise and the number of channels. To make the detector better adapt to the characteristics of SAR image, we train our detector from scratch without using the pre-trained weights on other optical dataset.

To evaluate the performance of our proposed anchor-free ship detection method, we adopt four evaluation metrics, i.e., the precision, recall, F1 score, and average precision (AP) [47]. Precision denotes the proportion of all detected ships that are ground truth ships, recall denotes the proportion of all ground truth ships that are detected, both of them are the most widely used metrics. F1 score is the comprehensive evaluation metric, which combines precision and recall. AP is the quality evaluation metric, which considers both precision and recall, then models the area under precision-recall (PR) curves. All precision and recall results in this article are acquired based on 0.3 confidence score threshold and 0.75 NMS threshold. We calculate AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l according to the evaluation protocol on COCO dataset [48]. AP_{50} is the AP at IoU = 0.5, AP_{75} is the AP at IoU = 0.75, AP_s, AP_m, and AP_l represent AP for small-scale, medium-scale, and large-scale ship targets, respectively. Moreover, since there is only one target for ship detection, mean average precision (mAP) for ship target detection is equal to AP_{50}.

B. Ablation Study

In this section, we analyzed the contributions of different modules in our method with ablation experiments, i.e., the key-point estimation module and channel attention module. To guarantee the validity of experimental results, all experiments are set with the same settings. The overall results including precision, recall, AP, $\text{AP}_{0.5}$, $\text{AP}_{0.75}$, AP_s, AP_m, and AP_l of the ablation experiments are shown in Table III. The baseline is the detection results of CenterNet with backbone DLA-34. It can be seen that the proposed method with both the key-point estimation module and the channel attention module achieves remarkable improvement. Compared with the baseline, we can see that the precision, recall, AP, $\text{AP}_{0.5}$, $\text{AP}_{0.75}$, AP_s, AP_m, and AP_l are 0.7%, 1.2%, 3.0%, 1.7%, 3.3%, 2.8%, 3.5%, and 6.7% higher than those evaluated on baseline, respectively. According to the improvement in precision and recall, we can clearly see that the contributions of the key-point estimation module and the channel attention module are in different ways. In the following parts, we will analyze the contributions of the key-point estimation module and the channel attention module, respectively.

1) Contributions of Key-Point Estimation Module: As mentioned in the previous sections, when the ship targets lie alongside and connect densely in the inshore area, the detection performance drops dramatically. So the key-point estimation module is proposed to eliminate the missing detection caused

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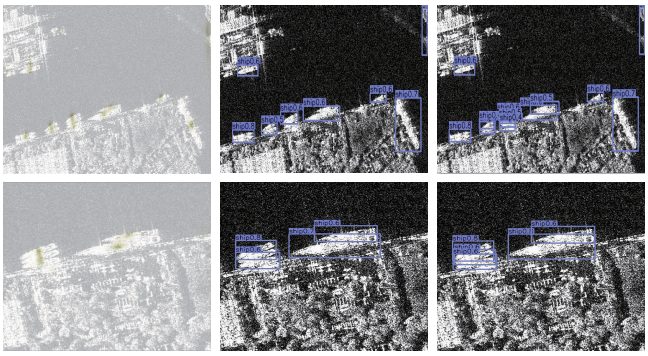
TABLE III
RESULTS OF ABLATION EXPERIMENTS

Key-point Estimation Module	Channel Attention Module	Precision	Recall	AP	$AP_{0.5}$	$AP_{0.75}$	AP_s	AP_m	AP_l
×	×	0.941	0.949	0.592	0.946	0.680	0.528	0.671	0.653
✓	×	0.941	**0.968**	0.607	0.960	0.692	0.540	0.686	0.695
×	✓	**0.953**	0.949	0.610	0.956	0.692	0.543	0.687	0.702
✓	✓	0.948	0.961	**0.622**	**0.963**	**0.713**	**0.556**	**0.706**	**0.720**

by the aforementioned issue. Some experiments are designed to evaluate the contributions of the key-point estimation module. As shown in Table III, the first and second rows represent the detection performance of the baseline and the proposed method only with the key-point estimation module, respectively. It can be calculated that recall increase by 1.9%, which illustrates the effect of the key-point estimation module on reducing the missing alarm rate. It can be observed that AP, $AP_{0.5}$, $AP_{0.75}$, AP_s, AP_m, and AP_l are 1.5%, 1.4%, 1.2%, 1.2%, 1.5%, and 4.2% higher than those evaluated on the baseline, respectively. Specifically, the key-point heat-maps generated by the fully convolutional network and the corresponding detection results are shown in Fig. 5(a) and (b). The luminance of the pixels represents the possibility of the predicted center points. We can see that almost all ship targets centers are marked with the light pixels corresponding to the peak key-points of the heat-map.

However, there are still a lot of ship targets missed detection. Hence, it is apparent that the peak key-points cannot match to each ship target precisely caused by the suppression of adjacent targets. Under such situations, it is of great significance to improve the key-point estimation of CenterNet to adapt to the emergency of the ship targets lying alongside and connecting densely in the inshore area. So we alleviate the suppression of the adjacent targets in process of extracting the peak key-points of the heat-map and reduce the undetected ship targets when they lying alongside and connecting densely in the inshore area. When the center points of different ship targets are in a close location, it prevents the peak key-points of different ship targets from being suppressed and matches them to their corresponding targets precisely. Some detection results without and with the key-point estimation module are shown in Fig. 5(b) and (c). It is obvious that numerous undetected ship targets in the detection results without key-point estimation module are reappeared in the detection results with key-point estimation module and the promotion of detection performance and the adaptability of ship targets dense area is prominent.

2) Contributions of Channel Attention Module: Due to that current detection methods are difficult to disentangle the background noise from ship targets, especially in complex scenes, the false alarm can occur frequently. We explore the channel attention module to improve the ability of the network on extracting the ship target features and suppress the background noise. As shown in Table III, the first and third rows represent the detection performance of the baseline and the proposed method only with the channel attention module, respectively. It can be calculated that precision increase by 1.2%, which illustrates the effect of the channel attention module on reducing the false alarm rate. In the meantime, it can be observed that AP, $AP_{0.5}$, $AP_{0.75}$, AP_s, AP_m, and AP_l are 1.8%, 1.0%, 1.2%, 1.5%, 1.6%, and 4.9% higher than those evaluated on baseline, respectively. As shown in Fig. 6(b) and (c) represent the detection results without and with the channel attention module. The contributions of the channel attention module in improving the ability of the network to disentangle background and noise can be illustrated distinctly.



(a) (b) (c)

Fig. 5. Key-point heat-map, detection result without and with the proposed key-point estimation module. (a) Key-point heat-map. (b) Detection results without the key-point estimation module. (c) Detection results with the key-point estimation module. Luminance of the pixels represent the possibility of center points.

Moreover, to reveal the result of the channel attention module intuitively, we implement the proposed method without and with the channel attention module in true SAR images in complex scenes. Then we hook the feature maps of them from the last convolutional layer and remap it to the corresponding input images. The visualization feature maps extracted without and with the channel attention module are shown in Fig. 7. There are more highlight features of ship targets and less attention to the background after combining the proposed detection method with the channel attention module. It is visible that the effect of the channel attention module is remarkable and significant.

C. Performance and Analysis

For the purpose of verifying the advantages in our method, YOLOv3, SSD-512, Faster R-CNN, FCOS, CenterNet, FBR-Net [15], and SSE-Net [16] are all implemented in the same dataset settings. MMdetection toolbox[1] is used to implement YOLOv3, SSD-512, Faster R-CNN, and FCOS. To ensure the fairness of the results, we use the same backbone in these ship target detection methods as much as possible. In addition,

[1]MMdetection is an open source target detection toolbox based on PyTorch, it is publicly available: https://github.com/open-mmlab/mmdetection

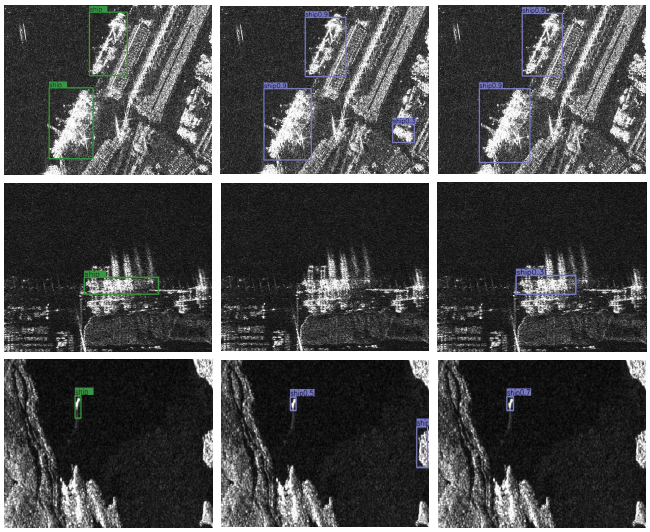
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TABLE IV
PERFORMANCE OF YOLOV3, SSD-512, FASTER R-CNN, FCOS, CENTERNET, FBR-NET, AND THE PROPOSED METHOD ON SSDD

Method	Backbone	Precision	Recall	F1	mAP
YOLOv3	DarkNet-53	0.926	0.934	0.930	0.949
SSD-512	VGG-16	0.891	0.908	0.899	0.942
Faster R-CNN	ResNet-50+FPN	0.944	0.910	0.927	0.909
FCOS	ResNet-50+FPN	0.881	0.946	0.912	0.936
CenterNet	DLA-34	0.941	0.949	0.945	0.946
FBR-Net [15]	ResNet-50+FPN	0.928	0.940	0.934	0.941
The proposed method	DLA-34	**0.948**	**0.961**	**0.954**	**0.963**

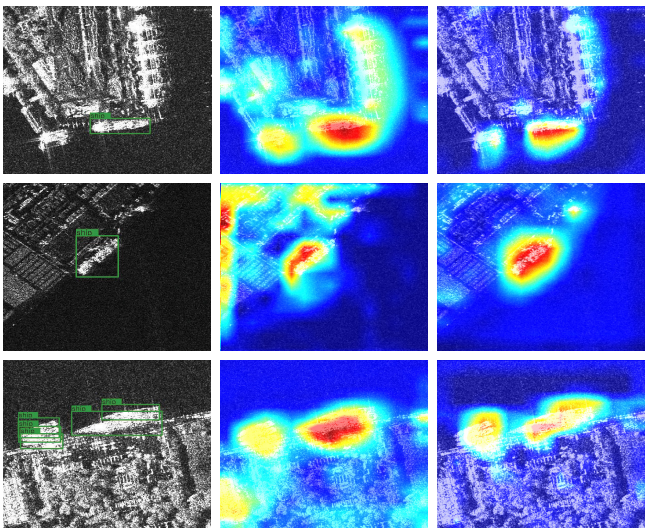
TABLE V
PERFORMANCE OF CENTERNET, SSE-NET, AND THE PROPOSED METHOD ON SAR-SHIP-DATASET

Method	AP	$AP_{0.5}$	$AP_{0.75}$	AP_s	AP_m	AP_l
CenterNet	0.544	0.924	0.583	0.483	0.626	0.722
SSE-Net [16]	0.609	0.947	**0.691**	0.547	**0.694**	0.739
The proposed method	**0.611**	**0.955**	0.678	**0.554**	0.675	**0.765**



(a) (b) (c)

Fig. 6. Detection results without and with the channel attention module. (a) Ground truth. (b) Detection results without the channel attention module. (c) Detection results with the channel attention module.



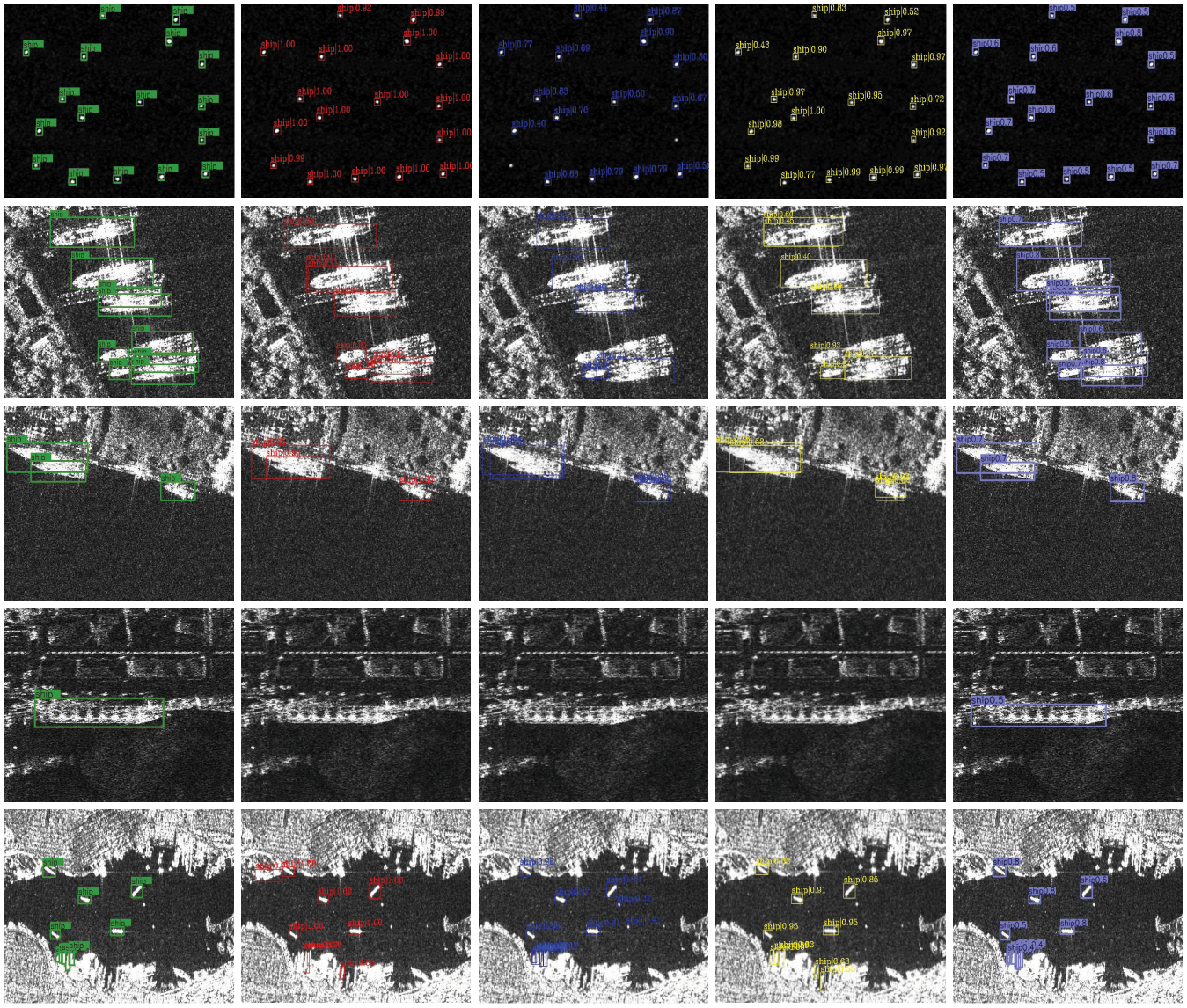
(a) (b) (c)

Fig. 7. Visualization features without and with the channel attention module. (a) Ground truth. (b) Feature maps without the channel attention module. (c) Feature maps with the channel attention module.

FPN is utilized to encourage the reuse of bottom features and add spatial information. Precision, recall, F1 score, and mAP on SSDD can be considered as the performance judgment of different detectors in Table IV. It is necessary to mention that because of the differences in the images between FBR-Net, SSE-Net and other detection methods test sets, the results of the evaluation may exist distinction. Through the comparison of precision, recall, F1 score, and mAP, our method is proven to achieve a higher detection accuracy with lower false alarm rate. Meanwhile, mAP of our method is 1.4%, 2.1%, 2.7%, and 5.4% higher than YOLOv3, SSD-512, FCOS, and Faster R-CNN. The detection performance of CenterNet, SSE-Net, and the proposed method on SAR-ship-dataset are shown in Table V. It can be observed that AP, $AP_{0.5}$, $AP_{0.75}$, AP_s, AP_m, and AP_l are 6.7%, 3.1%, 9.5%, 7.1%, 4.9%, and 4.3% higher than those evaluated on baseline, respectively. The proposed method has an obvious advantage with the mAP of 95.5%.

Fig. 8 are the results of our method and other detectors. We use different colors to illustrate the detection results, i.e., Faster R-CNN (red), SSD-512 (blue), YOLOv3 (yellow), and the proposed method (purple). Specifically, as can be seen from the first row, our method is more sensitive to the ship targets with small size and all undetected ship targets of other detection methods are detected. Besides, the second and third

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(*a*) (*b*) (*c*) (*d*) (*e*)

Fig. 8. Detection results of different detection methods. (a) Ground truth. (b) Detection results of Faster R-CNN. (c) Detection results of SSD-512. (d) Detection results of YOLOv3. (e) Detection results of the proposed method.

rows reveal that our method maintains a superior detection performance when dealing with the densely connected ship targets in the inshore area and the suppression of the adjacent ship targets is alleviated observably. Furthermore, the fourth and last rows illustrate the ability of the proposed method to disentangle background information and noise from the ship targets in complex scenes.

In summary, it can be analyzed from the experiment results that the proposed method processes a better detection performance with 96.3% mAP and 95.4% F1 score on SSDD, 95.5% mAP on SAR-ship-dataset. Moreover, the proposed method not only achieves a high detection accuracy when facing multiscale and dense ship targets but also performs a low false alarm rate when detecting the ship targets in the inshore area and with severe speckle noise. Furthermore, in our hardware platform and experimental settings, the detection speed of the proposed method is 26 frames per second (FPS) with the memory usage of 1189 MB. Therefore, our method is more practical in detecting multiscale and dense ship targets in complex scenes.

V. Conclusion

In this article, we propose an anchor-free SAR ship detection method toward multiscale and dense ship targets. The proposed method takes anchor-free detection method CenterNet as the framework, and has good adaptability to the size of multiscale ship targets. Moreover, the key-point estimation module is proposed to eliminate the missing alarm when detecting the densely distributed ships in the inshore area. Furthermore, the channel attention module is designed to suppress the background noise, and reduce the false alarm in complex scenes. A series of comparison experiments with the same dataset settings on different detection methods prove the advantages of the proposed method, especially in multiscale and dense ship targets detection.

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In the future, we will continually refine the proposed method and test it on other SAR images data with different distributions, including but not limited to SAR images data in different frequency bands. In the meantime, we will focus on the emerging issues in the field of SAR images ship target detection including but not limited to multiscale and dense ship targets detection in complex scenes.

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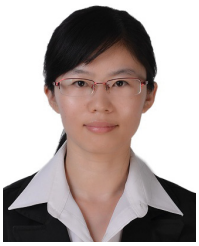
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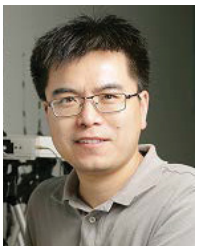
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